

CHAPTER 5

Hardware-Aware Co-Design Framework for Efficient TinyML Deployment on Ultra-Low- Power Edge Devices

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ABSTRACT

Resource-constrained edge environments define the operational limits of TinyML deployment. Memory capacity, computational capability, and energy availability remain tightly bounded. Conventional deep neural architectures exceed these limits. Practical adoption is therefore restricted. Existing optimization strategies operate in isolation. Neural architecture search, model compression, and runtime adaptation are applied independently. Coordinated efficiency is not achieved under such fragmented approaches. A hardware-aware co-design framework is introduced to address this limitation. Architecture synthesis, compression, and runtime scheduling are integrated within a closed-loop optimization process. Multi-objective hardware-aware NAS generates architectures under memory and energy constraints. Model complexity is reduced through quantization, structured pruning, and knowledge distillation. Parameter space is compressed. Computational demand decreases. Adaptive runtime scheduling regulates inference behaviour. System conditions such as battery level, computational load, and input complexity influence execution. Dynamic adjustment improves efficiency under varying conditions. Implementation is performed on ultra-low-power microcontroller platforms. System performance is evaluated across accuracy, memory footprint, and energy consumption. Resource usage is reduced. Predictive performance remains competitive. Experimental outcomes indicate that integrated co-design enables efficient deployment under constrained environments. The gap between theoretical optimization and practical implementation is reduced. Scalable and adaptive edge intelligence becomes achievable under this unified framework.

Keywords—TinyML, Edge Computing, Hardware-Aware Co-Design, Neural Architecture Search (NAS), Model Compression, Quantization, Pruning, Knowledge Distillation, Ultra-Low-Power Devices, Embedded Systems, Adaptive Runtime Scheduling, Energy-Efficient Inference

I. INTRODUCTION

Data-driven representation learning defines the capability of deep neural networks. Complex patterns are extracted from large-scale datasets. Performance gains are